



A year ago, the average price of a low-end board was about Rs 4,000. Today, the cheapest board—the Mercury PVM266AM-L—costs a meagre Rs 2,350 and goes up to an affordable Rs 4,900. With this board as your substrate, you can assemble a complete PC for as little as Rs 20,000. Will such a PC do for an average user? Will it make a good home PC? These are the questions we kept in mind while testing this category.

Take a quick look at the score board, and you'll see that three to four chipsets are predominantly used in this segment. The Intel 845 chipset-based motherboards were dominant—two thirds of the boards were based on it. The VIA P4M266A was also popular, and the 650 chipset from SiS was used on two boards. There was also an 865-based board stripped to the bare necessities.

Features

The low-end boards have decent features—six USB ports, an AGP slot, and onboard LAN. All the boards were based on Micro ATX form factor, and hence the component layout was quite cramped. The boards with an AGP slot were worse—the memory slot locks interfered with the AGP card. Boards from AOpen, ASUS, Gigabyte, Intel and Mercury displayed better component layout. The layout on the Jetway-P4MDUT was a complete mess, with capacitors in all the wrong places.

Most of the Intel chipset boards were based on the value 845GV chipset. This offers good features with acceptable performance. Except for three boards—Intel 845GVSR, HIS 53-1A and Jetway IGVM—all the others had onboard LAN. Most of the Intel-based boards supported DDR 333 MHz RAM, while the VIA P4M266 chipset boards supported DDR 266 MHz. The Asrock M266A had support for both DDR 266 MHz, as well as PC 133 SD-RAM—a boon if you want to upgrade at minimal costs. The three different boards based on the three different SiS chipsets, supported DDR 266 MHz to DDR 400 MHz RAM. Only two boards—Gigabyte's GA-8I865GVMK (based on 865 chipset), and their GA-8S648FX (based on the SiS 648FX chipset), supported a 800 MHz FSB CPU and DDR 400 MHz memory.

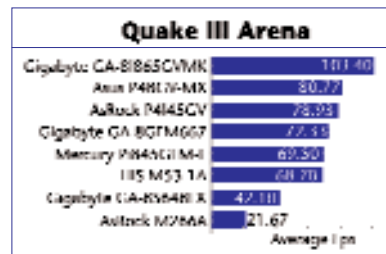
The 865 chipset board landed in this category only because of the lack of an AGP slot. Some boards such as those from Gigabyte, ASUS and AOpen did have BIOS overclocking options. The VIA and SiS chipsets scored over the Intel-based boards as they had 4X,

and in some cases 8X, AGP slots. However, a 4X slot isn't really much of a boon because most AGP cards available today are 8X-compatible; so a 4X card would only give you marginally better graphics than the onboard graphics sub-system.

As far as packaging was concerned, all the boards came with the standard pair of IDE cables for hard disk and floppy drives, rear panel plate and manuals. The AOpen board had just a quick start guide, with the complete manual on the CD—how they expect you to look up jumper settings before you assemble your computer is a complete mystery!



The ASRock M266A motherboard has a micro ATX form factor and hence comes with a cramped layout



Performance

The low-end category caters to those who require a PC for word processing, browsing, listening to MP3s and playing a few graphically unchallenging games. Price is the most important factor here, and we took into consideration that the total budget for a PC based on these motherboards shouldn't cross Rs 20,000. Also, the machine should also be able to last a few years without an upgrade.

The real-world test gauges how good a board is at day to day activities. In content creation, Gigabyte's GA-8I865GVMK scored a superb 27.4. This was due to the 865 chipset, and its high memory bandwidth. The board from Gigabyte—the GA-8S648FX—based on SiS 648FX chipset with similar specifications, came in second. The MSI and Jetway's IGVM tied for third spot with scores around the 23 mark—a good performance for boards based on 845 chipset.

In the Business Winstone benchmark, both the Gigabyte boards took the top honours—clearly indicating what the extra memory bandwidth can do for you. The Gigabyte GA-8GEM667, based on Intel 845GE chipset, came in third. The ASUS P4BGV-MX did a decent job as well. The lower memory bandwidth of the VIA chipset boards created a bottleneck for memory-intensive applications and tasks, putting office applications out of their range.

No prizes for guessing who encoded the fastest—it was the duo from Gigabyte again! The VIA chipset-based motherboards came in last, while the 845 chipset based ASUS and Gigabyte GA-8GEM667 managed to do the job fairly quickly.

In the synthetic test, especially the PC Mark benchmark, which yields a unified score, the same fact came to light. Gigabyte's GA-8I865GVMK yielded more points than any other board. The reason was the dual-channel implementation of memory on the 865 chipset. The VIA chipset based boards were a disappointment again, owing to completely outdated 266 MHz memory bus. In the Intel 845 clan, Gigabyte's slightly better 845GE chipset motherboard topped. All the boards were on par in the

